

TRIAD FINITE PLANNER

decides *WHAT* and *WHEN*

event time τ

1 of 14

meeting pose $\Pi(\tau)$

predicted, frozen

grasp g

1 of 32

route r

1 of 17

copied-state certification of every candidate through post-capture retreat

→ J_{global} → **exact finite argmin over the timing-admissible set**

the argmin exists ONLY in this layer



one committed plan $\xi^* = (\tau^*, g^*, r^*)$

TRIAD SUPERVISORY FSM

enforces *WHEN TO ACT* — and fails closed

live timing admission · prediction freshness (0.015 m / 0.120 rad) · one-shot commit
22 post-commit guards · clearance governor · bisecting safety filter · gripper interlock
capture → transfer → retreat · no replanning · no retiming · no reselection

any guard violation → fail-safe hold → Failure



target pose · body-frame reference velocity · posture · gripper

no event time, grasp, route or objective crosses this boundary

mc_rtc TASK / QP LAYER

realizes *HOW* — it does not select

one TransformTask (target pose, body-frame refVel, refAccel = 0)

posture task (per-joint weights and stiffnesses)

one kinematics constraint (joint position and velocity limits)

∅ **collisions: []**

∅ **contacts: []**

no collision constraint, no contacts



joint commands

SIMULATOR / FUTURE PHYSICAL ROBOT

mc_rtc_ticker today · Kinova Gen3 + Robotiq 2F-85 planned

all reported end-to-end results are simulation results — no physical handover result is claimed